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Basic Linux Networking Commands

This session (L6 S11) will cover essential networking commands that are critical for troubleshooting and managing network-related issues.

Basic networking commands:

Linux provides a variety of networking commands to manage and troubleshoot network connections.

Here are some basic and commonly used commands:

- a) ping
- b) ip
- c) ifconfig
- d) netstat
- e) ss

Command	Description
ping	Test connectivity to a remote host
ifconfig	Configure network interfaces (deprecated)
ip	Manage network interfaces, IP addresses, and routes
netstat	Display network connections and statistics (deprecated)
ss	Display socket statistics and network connections

Basic Linux Networking Commands

a) ping

- The ping command is used to test connectivity between two nodes on a networks.
- It works by sending ICMP echo request packets and waiting for a reply.

Syntax:

ping [options] destination

Examples:

- **Ping a hostname**

ping www.google.com

- **Ping an IP address**

ping 8.8.8.8

- **Send a specific number of packets**

ping -c 4 www.google.com

- **Specify packet size**

ping -s 56 www.google.com

Basic Linux Networking Commands

b) ip

- The ip command manages various network-related configurations such as IP addresses, interfaces, and routing tables.

Syntax:

```
ip [object] [command] [options]
```

Examples:

- **Display all network interfaces and their IP addresses**

```
ip addr show
```

- **Display detailed information about a specific interface**

```
ip addr show eth0
```

- **Assign an IP address to an interface**

```
sudo ip addr add 192.168.1.100/24 dev eth0
```

- **Remove an IP address from an interface**

```
sudo ip addr del 192.168.1.100/24 dev eth0
```

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Basic Linux Networking Commands

- **Bring an interface up**

sudo ip link set eth0 up

- **Bring an interface down**

sudo ip link set eth0 down

- **Display the routing table**

ip route show

- **Add a static route**

sudo ip route add 192.168.2.0/24 via 192.168.1.1

- **Delete a static route**

sudo ip route del 192.168.2.0/24

```
[tecmint@tecmint]$ sudo ip addr flush eth2
[tecmint@tecmint]$ ip -j -p addr show eth2
[ {
        "ifindex": 13,
        "link_index": 14,
        "ifname": "eth2",
        "flags": [ "BROADCAST","MULTICAST","UP","LOWER_UP" ],
        "mtu": 1500,
        "qdisc": "noqueue",
        "operstate": "UP",
        "group": "default",
        "link_type": "ether",
        "address": "02:42:ac:13:01:02",
        "broadcast": "ff:ff:ff:ff:ff:ff",
        "link_netnsid": 0,
        "addr_info": [ ]
    } ]
[tecmint@tecmint]$
```

Basic Linux Networking Commands

c) ifconfig

- It's command is used to configure network interfaces.
- But it is considered deprecated and replaced by the ip command in modern Linux distributions.

Syntax:

ifconfig [interface] [options]

Examples:

- **Display all network interfaces**

ifconfig

- **Display information about a specific interface**

ifconfig eth0

- **Assign an IP address to an interface**

sudo ifconfig eth0 192.168.1.100

- **Bring an interface up**

sudo ifconfig eth0 up

- **Bring an interface down**

sudo ifconfig eth0 down

Basic Linux Networking Commands

d) netstat

- The netstat command provides information about network connections, routing tables, and interface statistics.
- It is considered deprecated and replaced by the ss command in modern Linux distributions.

Syntax:

netstat [options]

Examples:

- **Display all active connections**

netstat

- **Display all listening ports**

netstat -l

- **Display TCP connections**

netstat -t

- **Display UDP connections**

netstat -u

- **Display routing table**

netstat -r

- **Display network interface statistics**

netstat -i

Basic Linux Networking Commands

e) ss

- The ss command is used for displaying socket statistics and is a modern replacement for netstat.
- The ss command is used to display socket statistics, and it’s a modern replacement for netstat.

Syntax:

ss [options]

Examples:

- **Display all open network connections**

ss -a

- **Display all listening ports**

ss -l

- **Display all TCP connections**

ss -t

- **Display all UDP connections**

ss -u

- **Display network statistics**

ss -s

- **Display processes using network connections**

ss -p